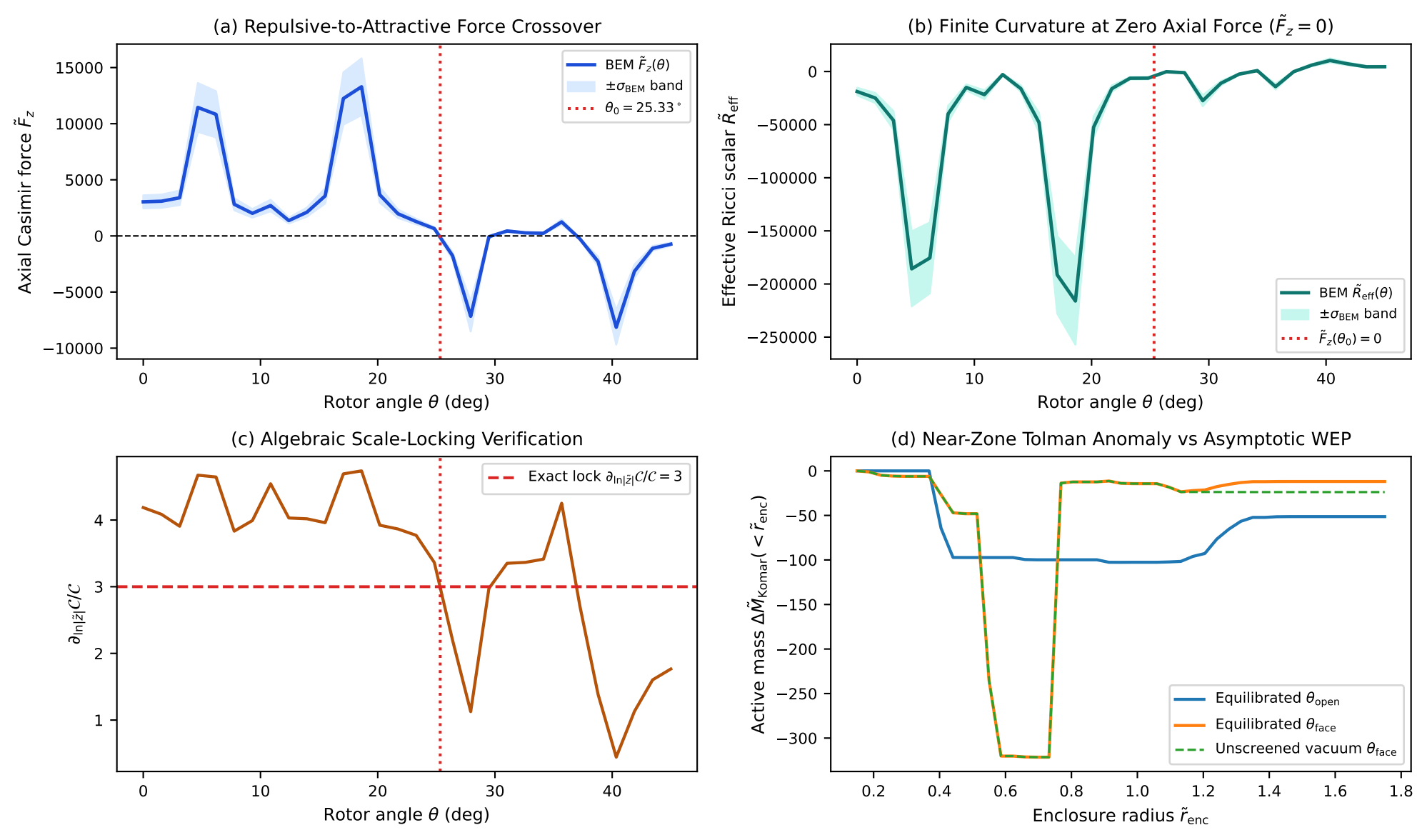
(a) Repulsive-to-Attractive Force Crossover
Axial Casimir force $\tilde{F}_z$
Rotor angle θ (deg)
BEM $\tilde{F}_z(\theta)$
$\pm\sigma_{\mathrm{BEM}}$ band
$\theta_0 = 25.33^\circ$
(b) Finite Curvature at Zero Axial Force ($\tilde{F}_z = 0$)
Effective Ricci scalar $\tilde{R}_{\mathrm{eff}}$
Rotor angle θ (deg)
BEM $\tilde{R}_{\mathrm{eff}}(\theta)$
$\pm\sigma_{\mathrm{BEM}}$ band
$\tilde{F}_z(\theta_0) = 0$
(c) Algebraic Scale-Locking Verification
$\partial_{\ln|\tilde{z}|}\mathcal{C}/\mathcal{C}$
Rotor angle θ (deg)
Exact lock $\partial_{\ln|\tilde{z}|}\mathcal{C}/\mathcal{C} = 3$
(d) Near-Zone Tolman Anomaly vs Asymptotic WEP
Active mass $\Delta\tilde{M}_{\mathrm{Komar}}(< \tilde{r}_{\mathrm{enc}})$
Enclosure radius $\tilde{r}_{\mathrm{enc}}$
Equilibrated θ_{open}
Equilibrated θ_{face}
Unscreened vacuum θ_{face}